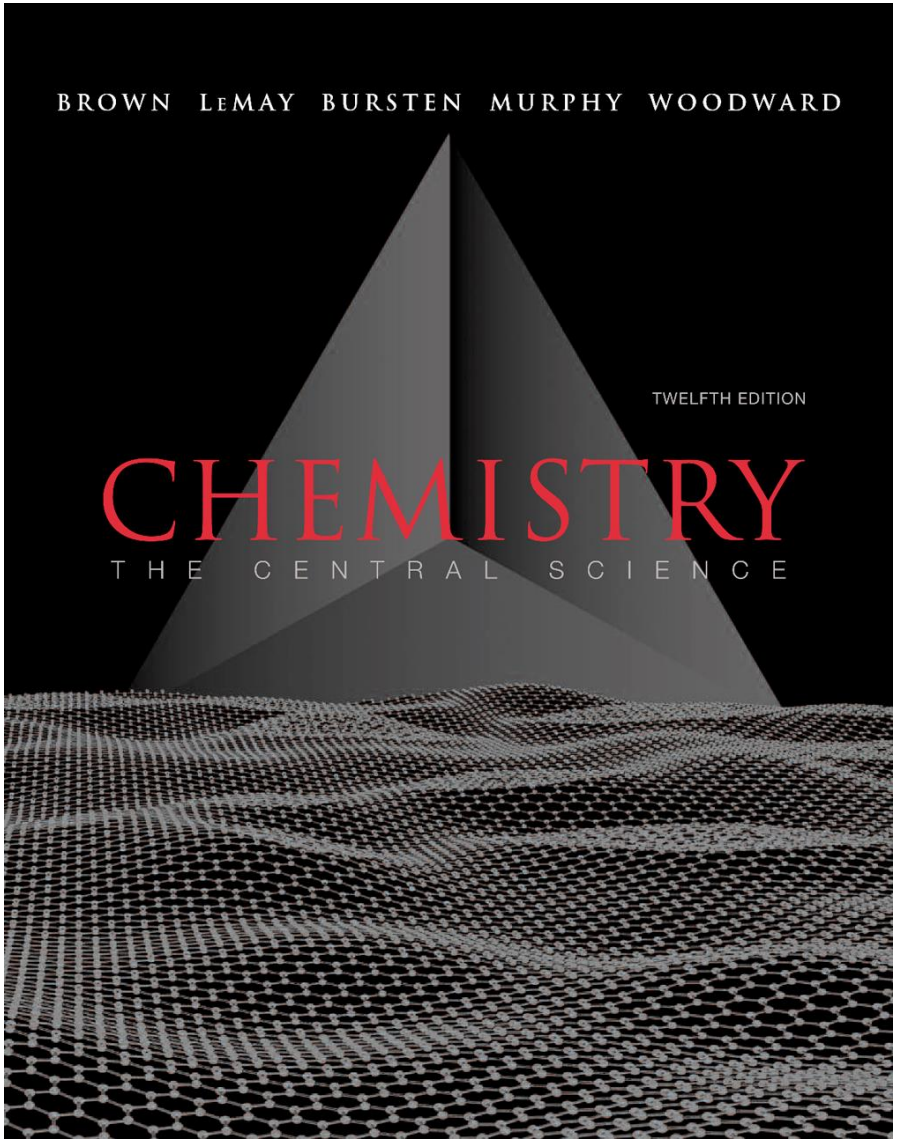


Lecture Presentation

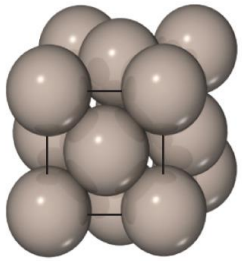
Chapter 12

Solids and Modern Materials

John D. Bookstaver
St. Charles Community College
Cottleville, MO

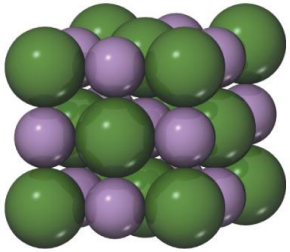
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Bonding in Solids



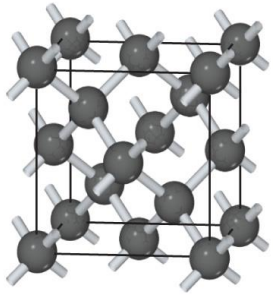
Metallic solids

Extended networks of atoms held together by metallic bonding (Cu, Fe)



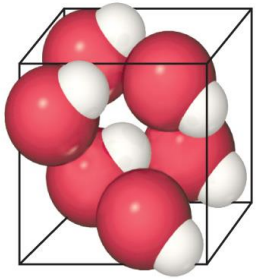
Ionic solids

Extended networks of ions held together by ion–ion interactions (NaCl, MgO)



Covalent-network solids

Extended networks of atoms held together by covalent bonds (C, Si)



Molecular solids

Discrete molecules held together by intermolecular forces (HBr, H_2O)

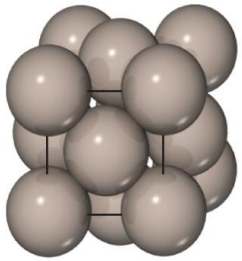
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- There are four general types of solids.
- **Metallic solids** share a network of highly delocalized electrons.
- **Ionic solids** are sets of cations and anions mutually attracted to one another.

Modern Materials

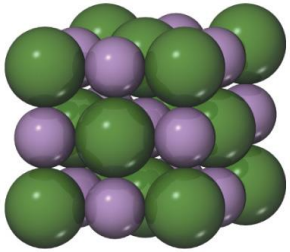
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Bonding in Solids



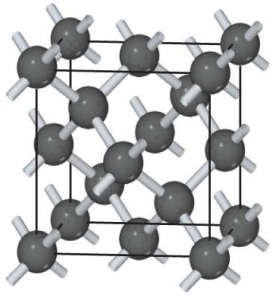
Metallic solids

Extended networks of atoms held together by metallic bonding (Cu, Fe)



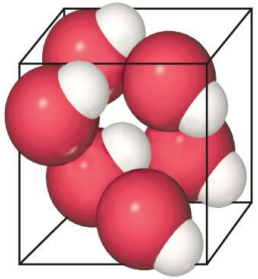
Ionic solids

Extended networks of ions held together by ion–ion interactions (NaCl, MgO)



Covalent-network solids

Extended networks of atoms held together by covalent bonds (C, Si)



Molecular solids

Discrete molecules held together by intermolecular forces (HBr, H_2O)

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- **Covalent-network solids** are joined by an extensive network of covalent bonds.
- **Molecular solids** are discrete molecules that are linked to one another only by van der Waals forces.

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Bonding in Solids

- In crystalline solids atoms are arranged in a very regular pattern.
- Amorphous solids are characterized by a distinct lack of order in the arrangement of atoms.



Iron pyrite (FeS_2), a crystalline solid



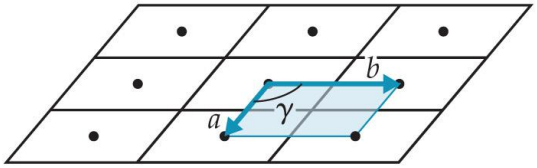
Obsidian (typically $KAlSi_3O_8$), an amorphous solid

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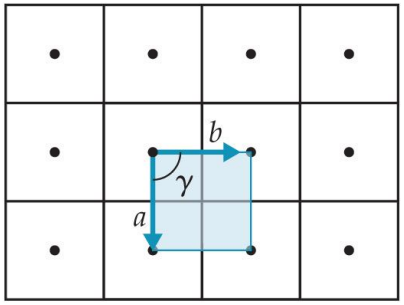
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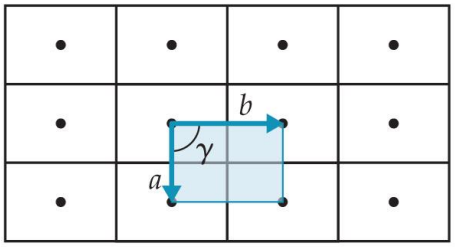
Crystal Lattices



Oblique lattice ($a \neq b$, γ = arbitrary)

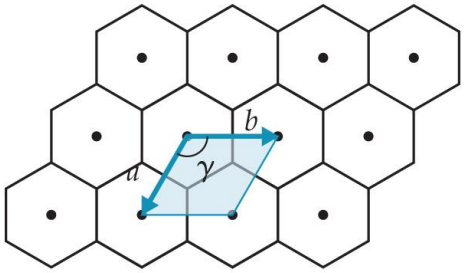


Square lattice ($a = b$, $\gamma = 90°$)



Rectangular lattice ($a \neq b$, $\gamma = 90°$)

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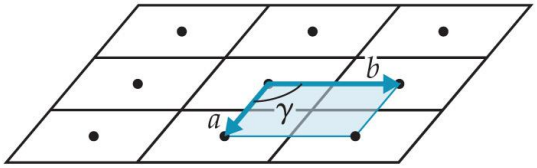
Hexagonal lattice ($a = b$, $\gamma = 120°$)

One can deduce the pattern in a crystalline solid by thinking of the substance as a lattice of repeating shapes formed by the atoms in the crystal.

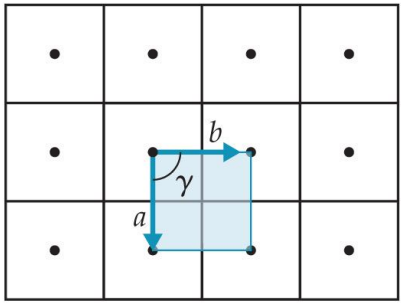
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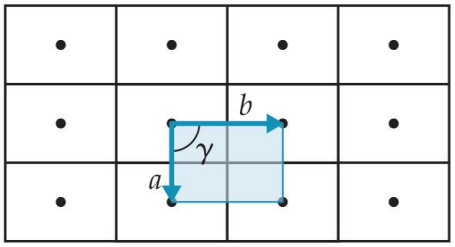
Crystal Lattices



Oblique lattice ($a \neq b$, γ = arbitrary)

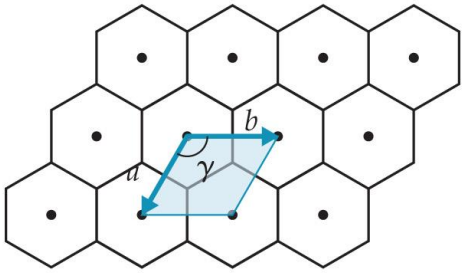


Square lattice ($a = b$, $\gamma = 90°$)



Rectangular lattice ($a \neq b$, $\gamma = 90°$)

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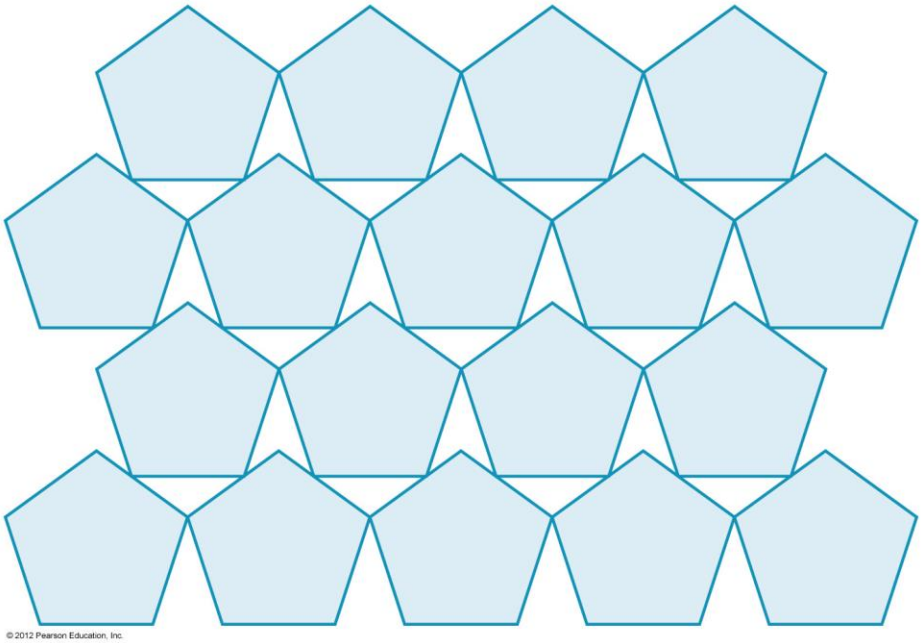
Hexagonal lattice ($a = b$, $\gamma = 120°$)

The individual shapes of the lattice, then, form "tiles," or **unit cells**, that must fill the entire space of the substance.

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Crystal Lattices



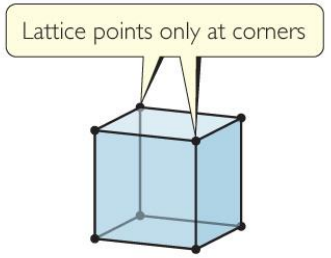
- There are seven basic three-dimensional lattices:
  - Cubic
  - Tetragonal
  - Orthorhombic
  - Rhombohedral
  - Hexagonal
  - Monoclinic
  - Triclinic

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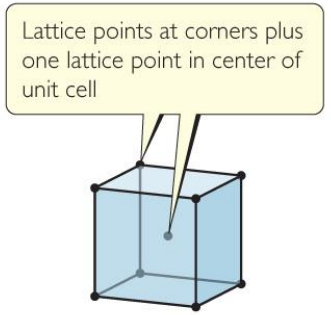
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Crystal Lattices

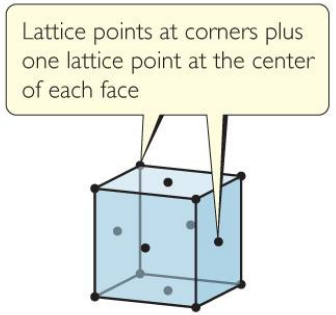
Within each major lattice type, additional types are generated by placing lattice points in the center of the unit cell or on the faces of the unit cell.



Primitive cubic lattice



Body-centered cubic lattice



Face-centered cubic lattice

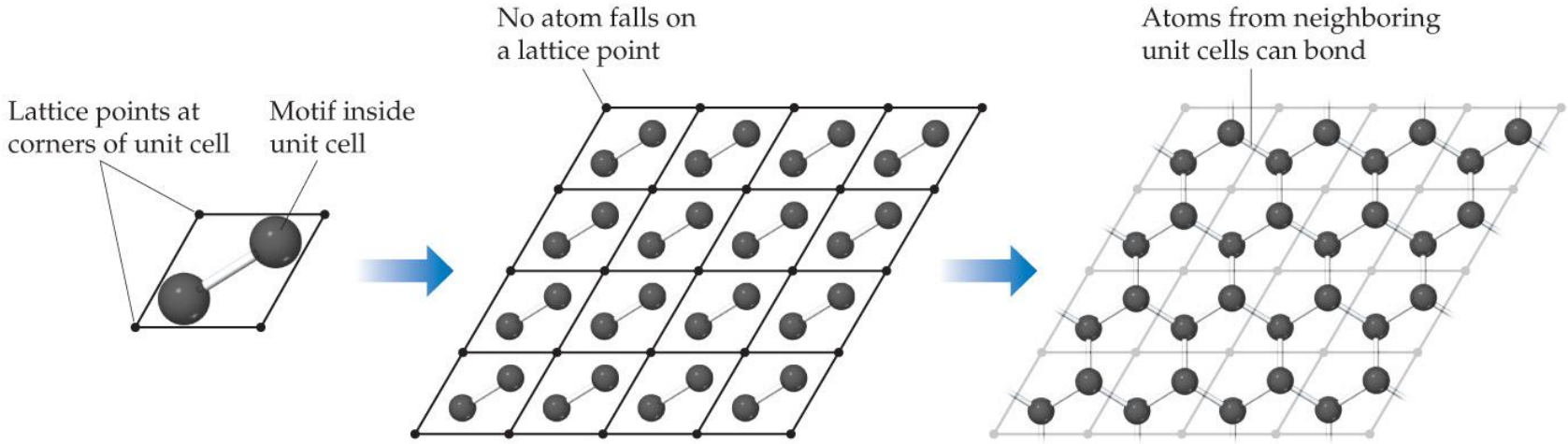
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Crystal Lattices

Once one places atoms within a unit cell, the structure of the compound can be seen by bonding the atoms to one another across unit cells.

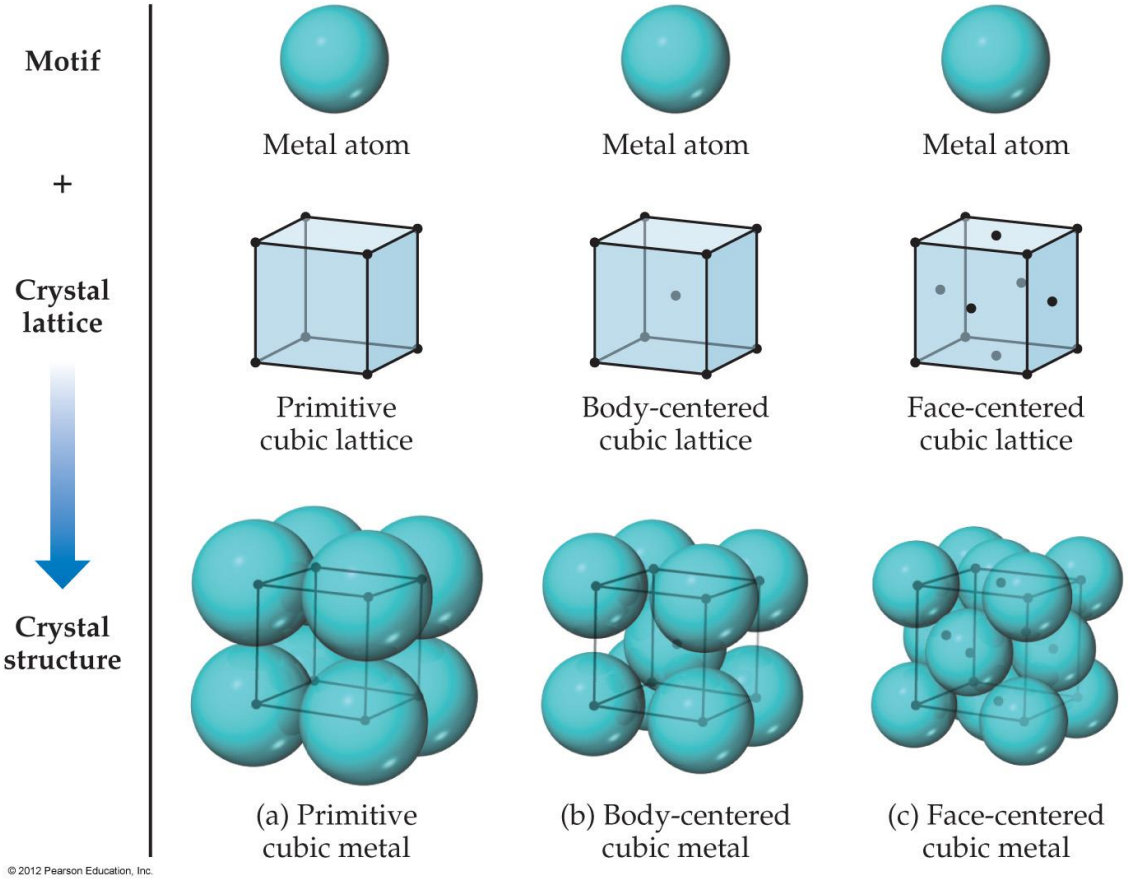


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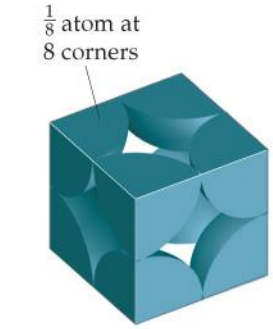
Metallic Structure



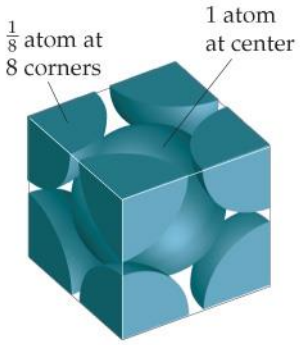
The structures of many metals conform to one of the cubic unit cells.

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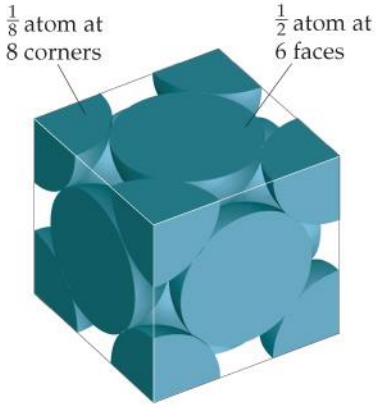
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(a) Primitive cubic metal
1 atom per unit cell



(b) Body-centered cubic metal
2 atoms per unit cell



(c) Face-centered cubic metal
4 atoms per unit cell

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Cubic Structures

One can determine how many atoms are within each unit cell which lattice points the atoms occupy.

TABLE 12.1 • Fraction of Any Atom as a Function of Location Within the Unit Cell*

Atom Location	Number of Unit Cells Sharing Atom	Fraction of Atom Within Unit Cell
Corner	8	1/8 or 12.5%
Edge	4	1/4 or 25%
Face	2	1/2 or 50%
Anywhere else	1	1 or 100%

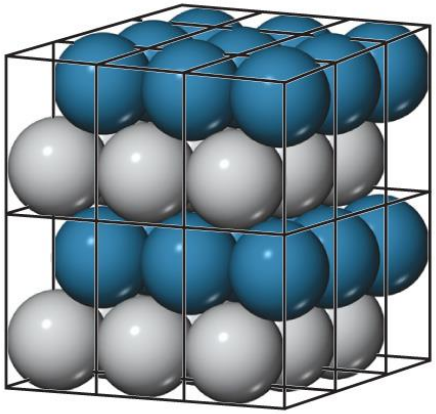
*It is only the position of the center of the atom that matters. Atoms that reside near the boundary of the unit cell but not on a corner, edge, or face are counted as residing 100% within the unit cell.

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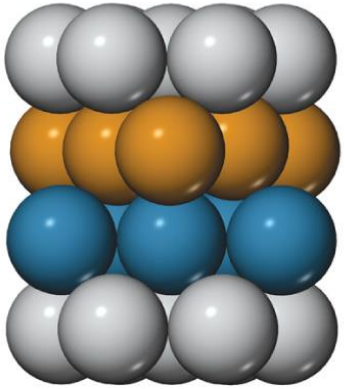
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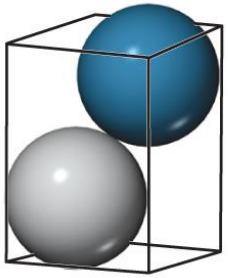
Close Packing



Side view

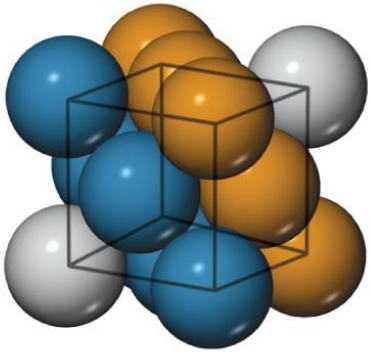


Side view



Unit cell view

(a) Hexagonal close-packed metal



Unit cell view

(b) Cubic close-packed metal

The atoms in a crystal pack as close together as they can based on the respective sizes of the atoms.

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Alloys

- **Alloys** are combinations of two or more elements, the majority of which are metals.
- Adding a second (or third) element changes the properties of the mixture to suit different purposes.

TABLE 12.2 • Some Common Alloys

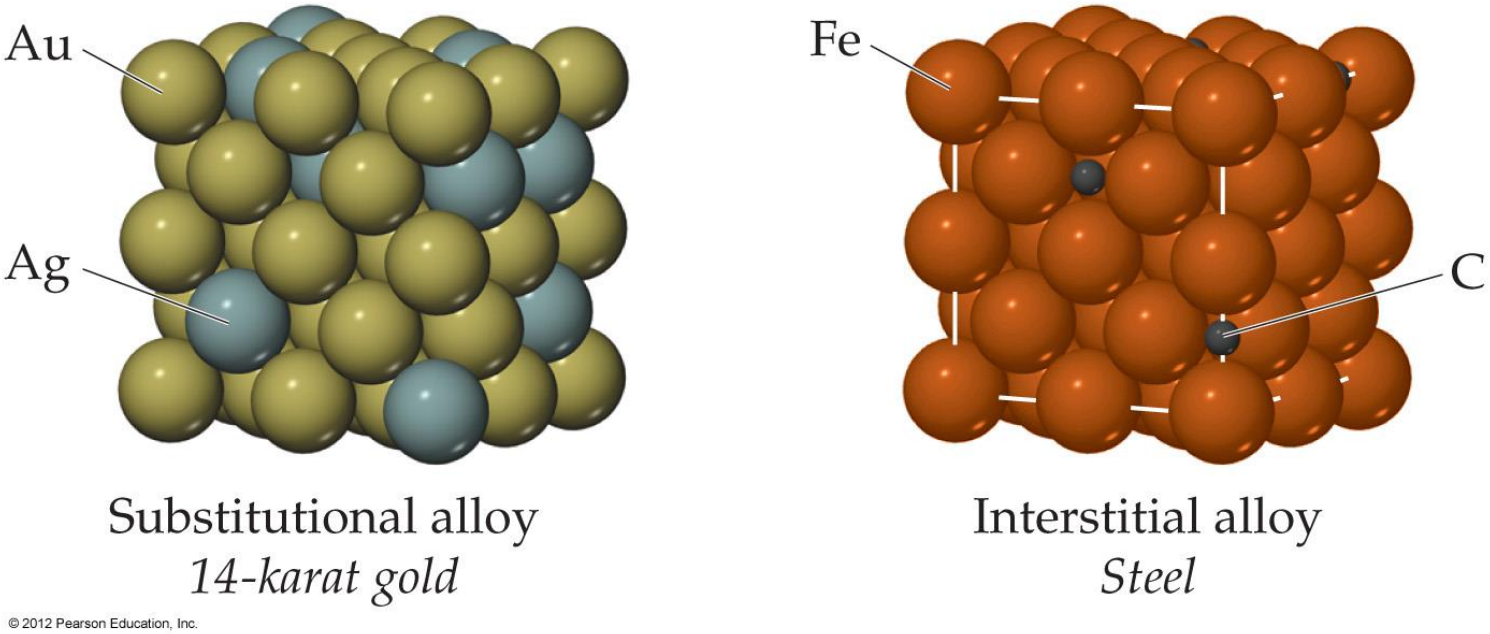
Name	Primary Element	Typical Composition (by Mass)	Properties	Uses
Wood's metal	Bismuth	50% Bi, 25% Pb, 12.5% Sn, 12.5% Cd	Low melting point (70 °C)	Fuse plugs, automatic sprinklers
Yellow brass	Copper	67% Cu, 33% Zn	Ductile, takes polish	Hardware items
Bronze	Copper	88% Cu, 12% Sn	Tough and chemically stable in dry air	Important alloy for early civilizations
Stainless steel	Iron	80.6% Fe, 0.4% C, 18% Cr, 1% Ni	Resists corrosion	Tableware, surgical instruments
Plumber's solder	Lead	67% Pb, 33% Sn	Low melting point (275 °C)	Soldering joints
Sterling silver	Silver	92.5% Ag, 7.5% Cu	Bright surface	Tableware
Dental amalgam	Silver	70% Ag, 18% Sn, 10% Cu, 2% Hg	Easily worked	Dental fillings

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Alloys



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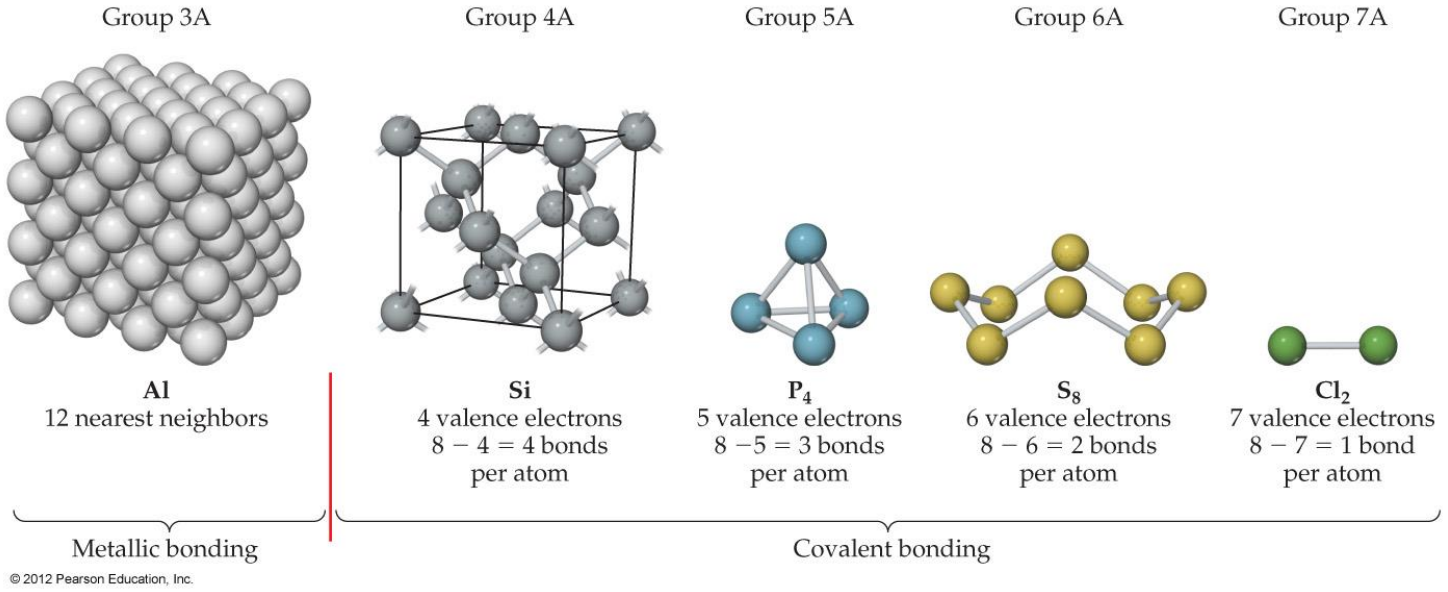
- In **substitutional alloys,** a second element takes the place of a metal atom.
- In **interstitial alloys,** a second element fills a space in the lattice of metal atoms.

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Metallic Bonding

- In elemental samples of nonmetals and metalloids, atoms generally bond to each other covalently.
- Metals, however, have a dearth of valence electrons; instead, they form large groups of atoms that share electrons among them.

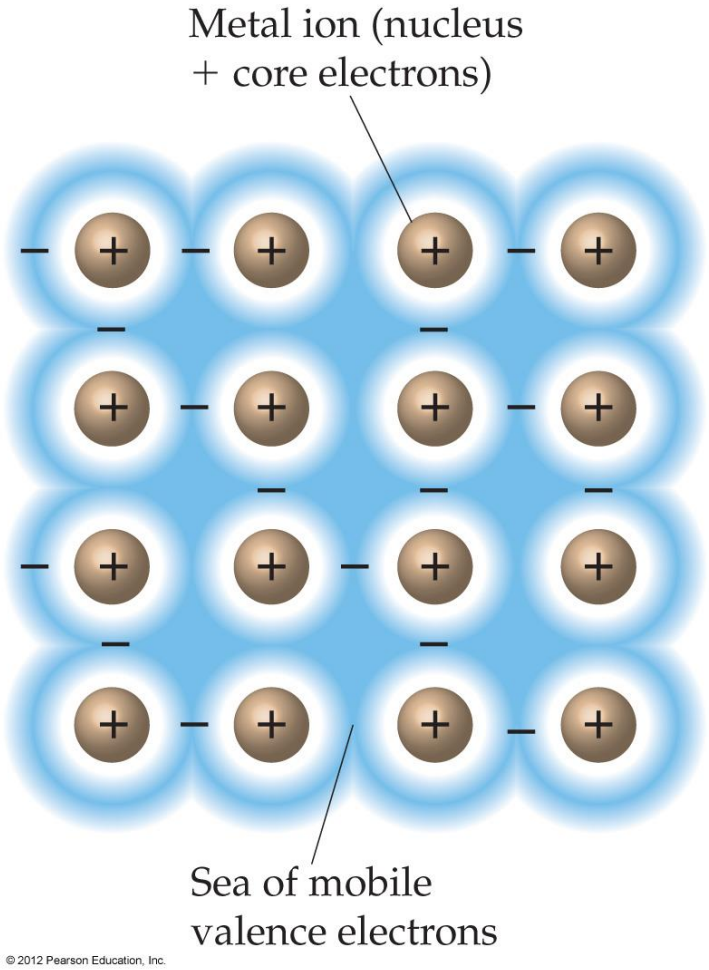


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Metallic Bonding

- One can think of a metal, therefore, as a group of cations suspended in a sea of electrons.
- The electrical and thermal conductivity, ductility, and malleability of metals is explained by this model.

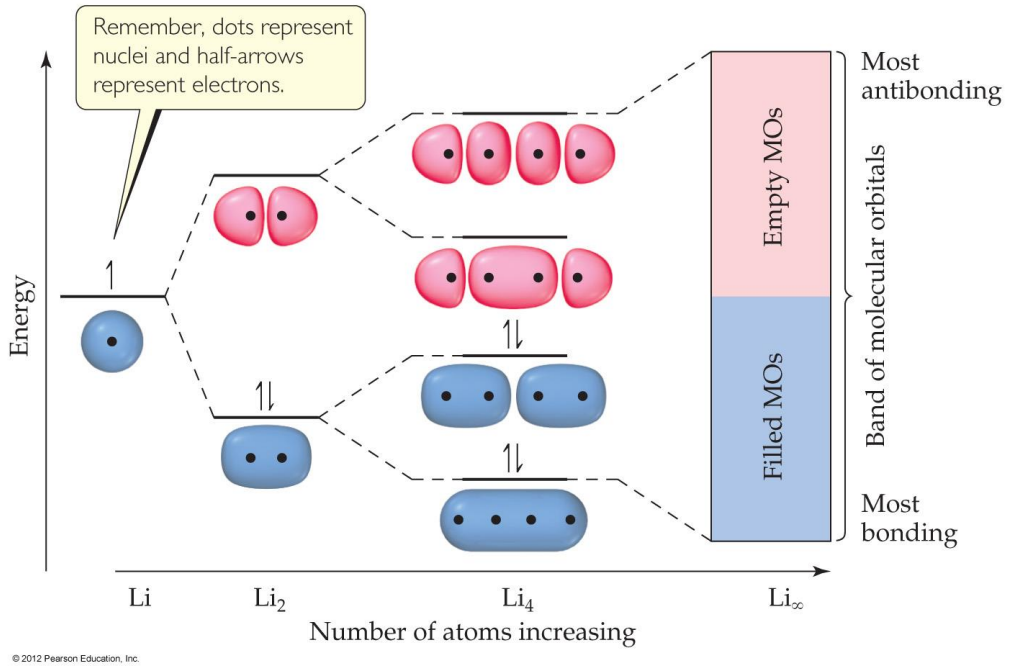


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A Molecular-Orbital Approach

As the number of atoms in a chain increases, the energy gap between molecular orbitals (MOs) essentially disappears, and continuous bands of energy states result.

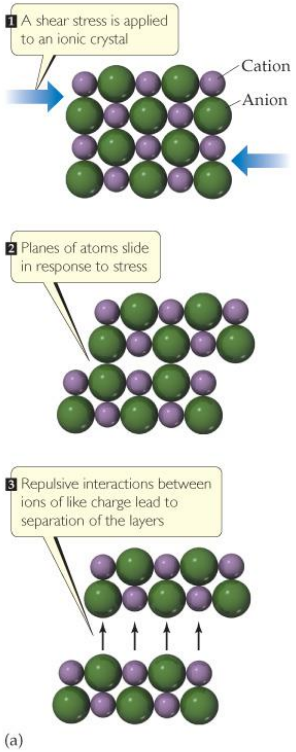


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Ionic Solids

- In ionic solids, the lattice comprises alternately charged ions.
- Ionic solids have very high melting and boiling points and are quintessential crystals.



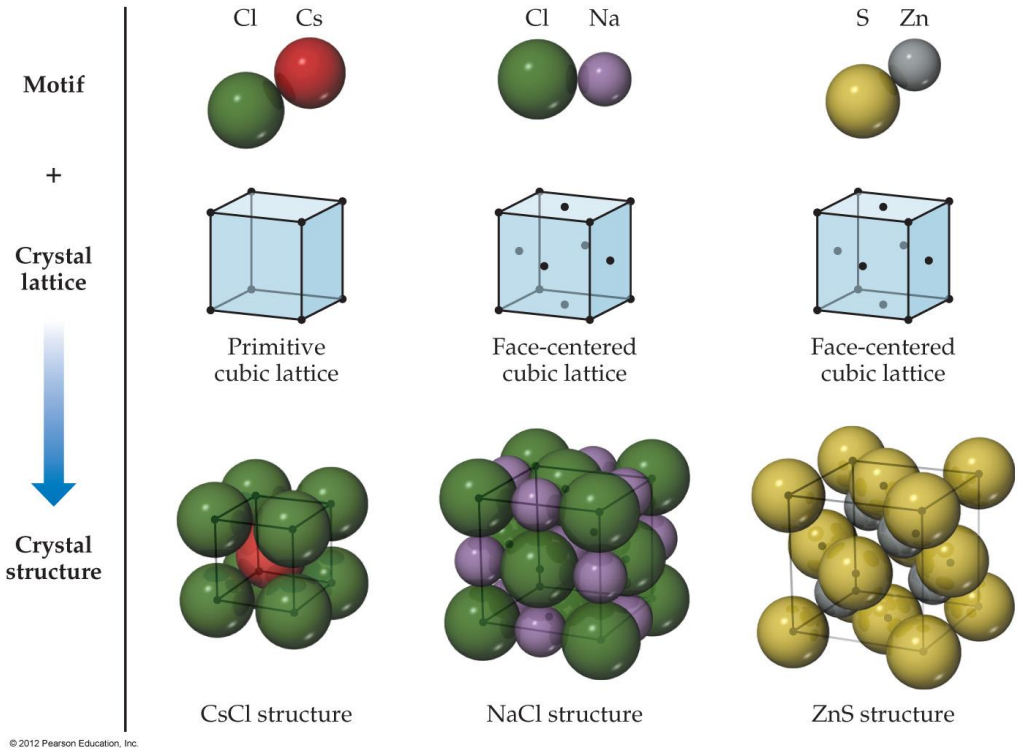

(b)

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Ionic Solids



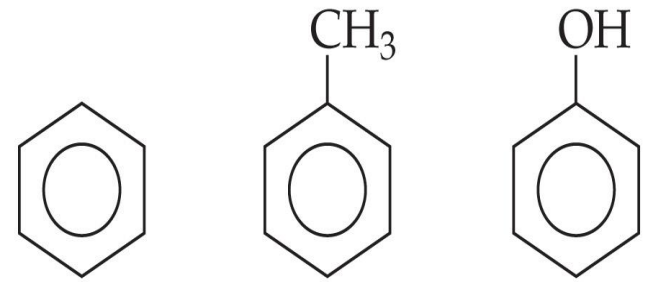
The different-sized ions in an ionic compound minimize the distance between oppositely charged ions while keeping like-charged ions away from each other.

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Molecular Solids

- The physical properties of molecular solids are governed by van der Waals forces.
- The individual units of these solids are discrete molecules.



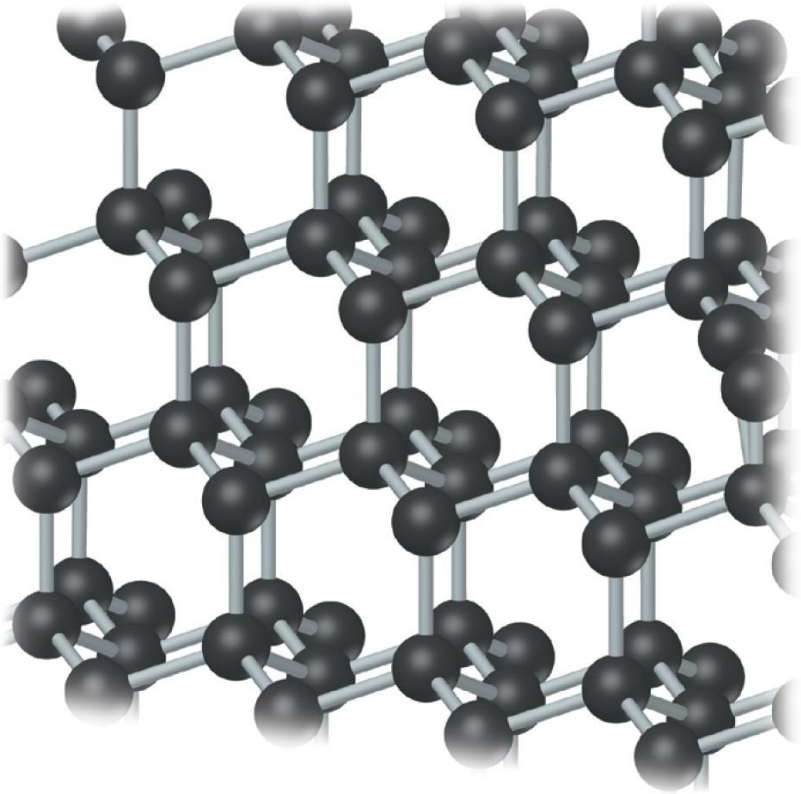
	Benzene	Toluene	Phenol
Melting point (°C)	5	−95	43
Boiling point (°C)	80	111	182

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Covalent-Network and Molecular Solids



(a) Diamond

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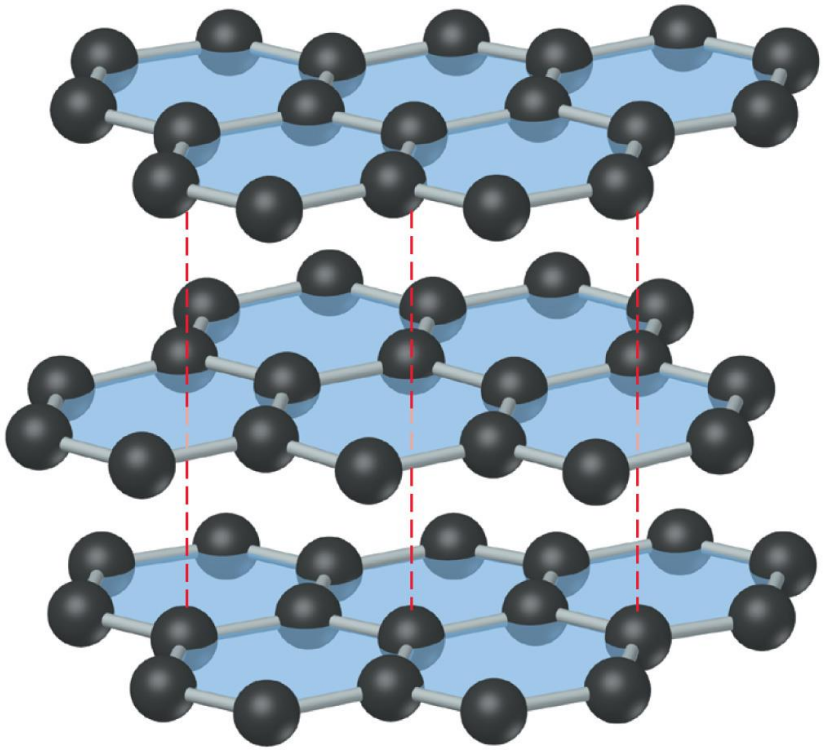
- Diamonds are an example of a covalent-network solid, in which atoms are covalently bonded to each other.
  - They tend to be hard and have high melting points.

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Covalent-Network and Molecular Solids

- Graphite is an example of a molecular solid, in which atoms are held together with van der Waals forces.
  - They tend to be softer and have lower melting points.



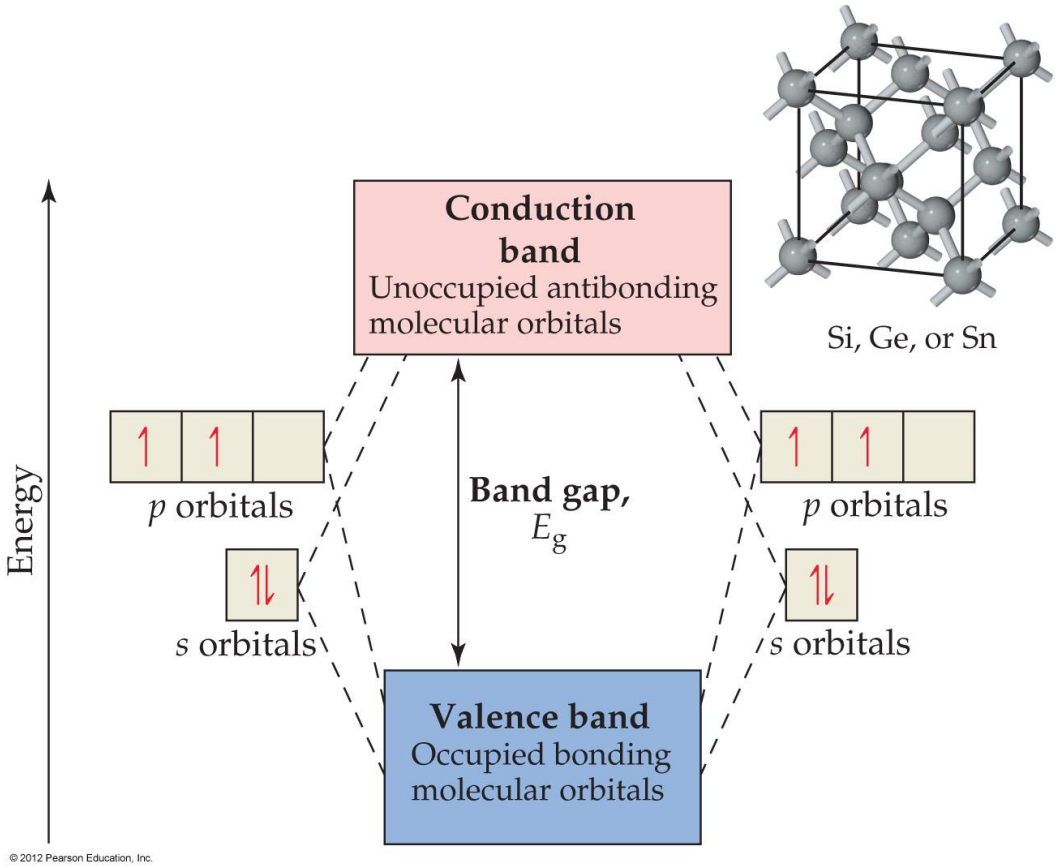
(b) Graphite

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Semiconductors

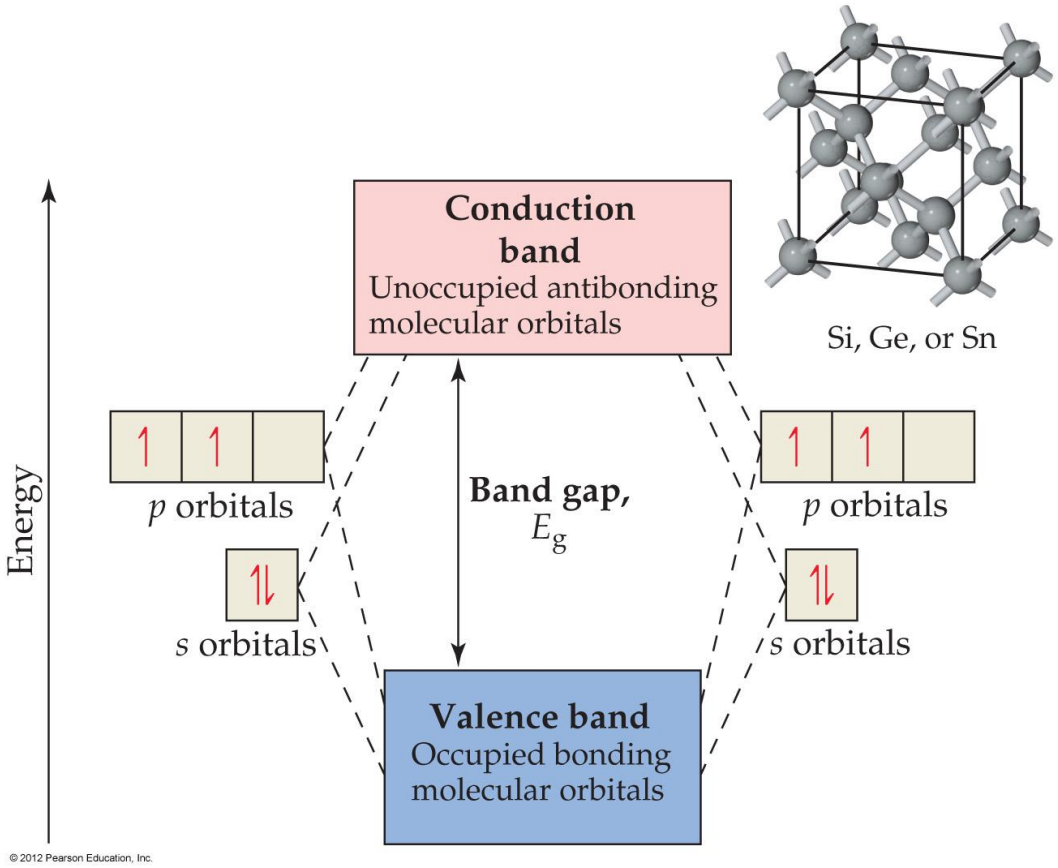


In the closely packed molecular orbitals in these compounds, there is a gap between the occupied MOs (valence band) and the unoccupied ones (conduction band).

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Semiconductors



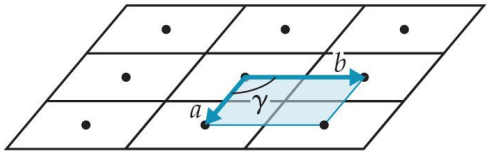
These Group IVA elements have gaps between their valence and conducting bands of 0.08 to 3.05 eV (7 to 300 kJ/mol).

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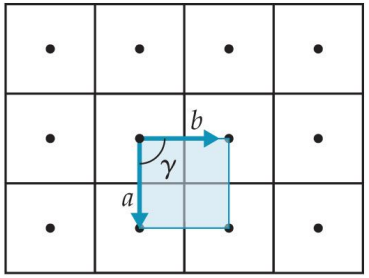
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Semiconductors

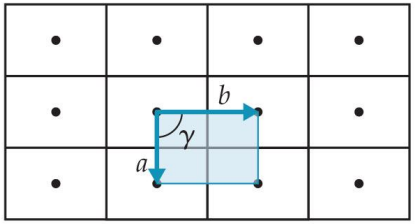
- Among elements, only Group IVA, all of which have 4 valence electrons, are semiconductors.
- Inorganic semiconductors (like GaAs) tend to have an average of 4 valence electrons (3 for Ga, 5 for As).



Oblique lattice ($a \neq b$, γ = arbitrary)

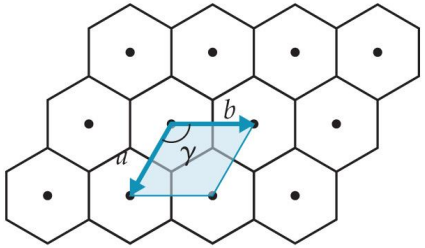


Square lattice ($a = b$, $\gamma = 90°$)



Rectangular lattice ($a \neq b$, $\gamma = 90°$)

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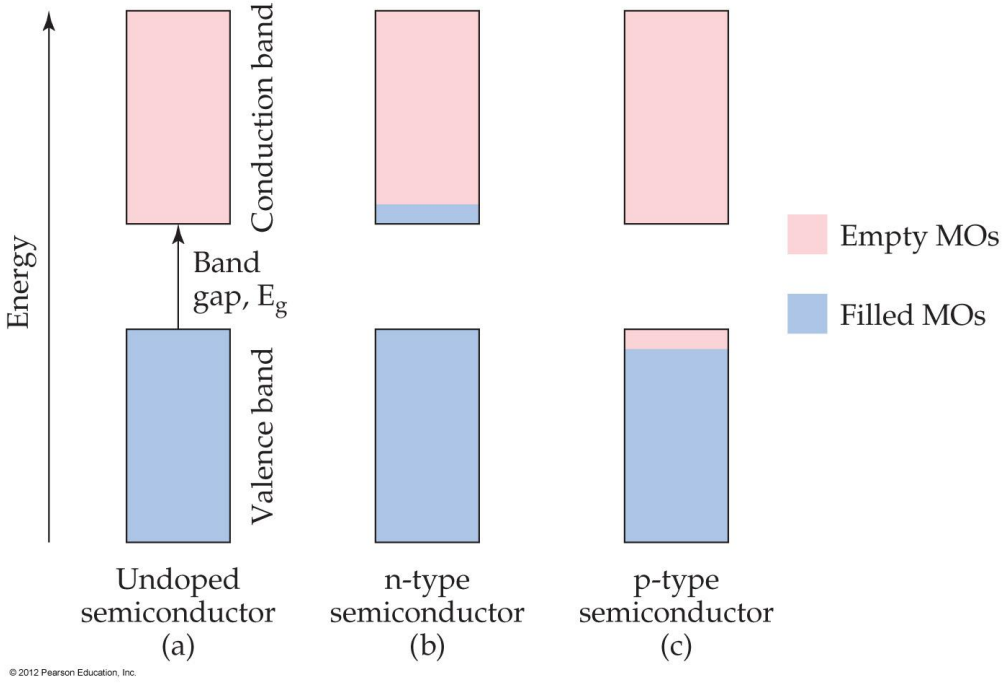
Hexagonal lattice ($a = b$, $\gamma = 120°$)

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Doping

By introducing very small amounts of impurities that have more (*n*-type) or fewer (*p*-type) valence electrons, one can increase the conductivity of a semiconductor.

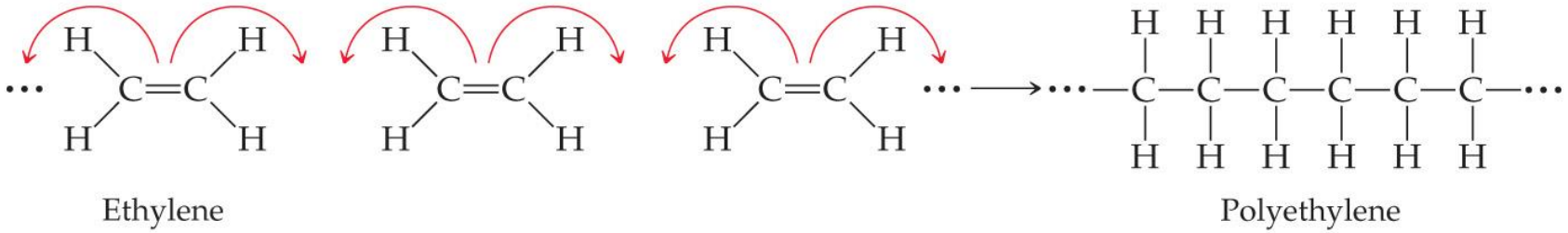


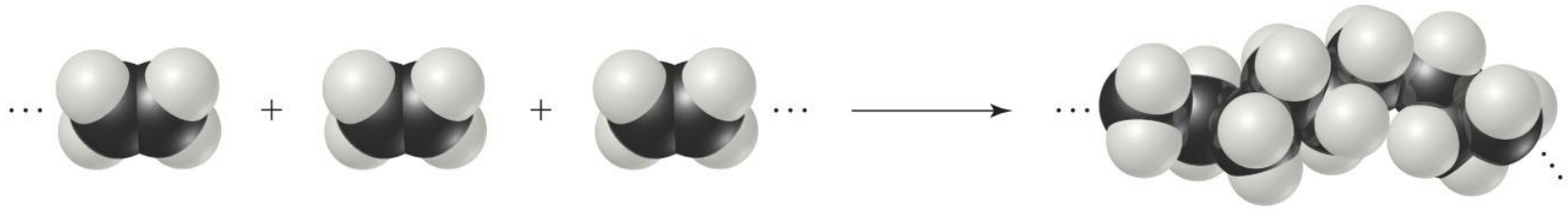
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Polymers

Polymers are molecules of high molecular mass made by sequentially bonding repeating units called **monomers**.





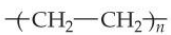
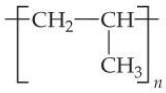
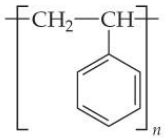
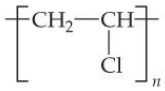
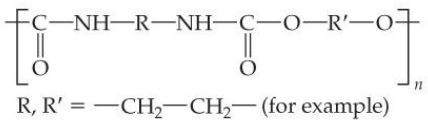
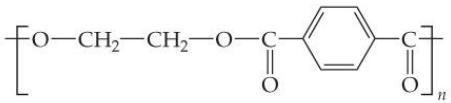
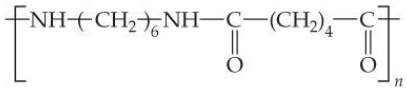
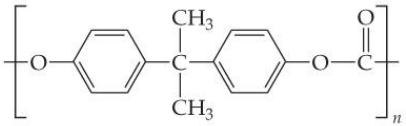
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Some Common Polymers

TABLE 12.5 • Polymers of Commercial Importance

Polymer	Structure	Uses
Addition Polymers		
Polyethylene		Films, packaging, bottles
Polypropylene		Kitchenware, fibers, appliances
Polystyrene		Packaging, disposable food containers, insulation
Polyvinyl chloride (PVC)		Pipe fittings, clear film for meat packaging
Condensation Polymers		
Polyurethane	 R, R′ = —CH_2—CH_2— (for example)	"Foam" furniture stuffing, spray-on insulation, automotive parts, footwear, water-protective coatings
Polyethylene terephthalate (a polyester)		Tire cord, magnetic tape, apparel, soft-drink bottles
Nylon 6,6		Home furnishings, apparel, carpet, fishing line, toothbrush bristles
Polycarbonate		Shatterproof eyeglass lenses, CDs, DVDs, bulletproof windows, greenhouses

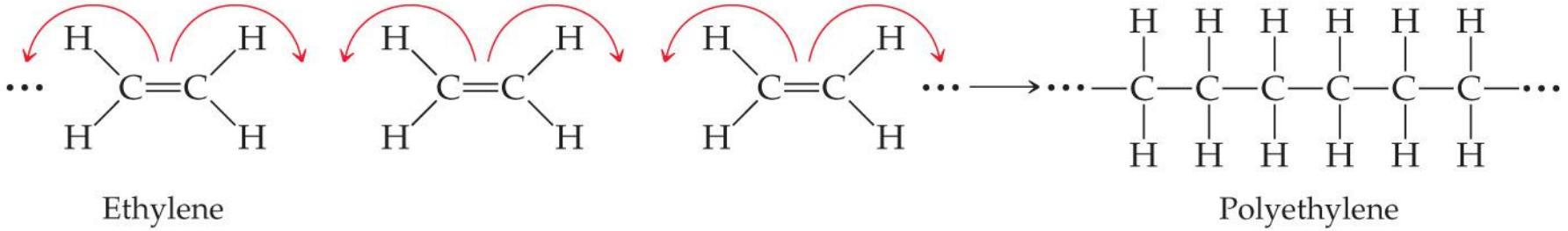
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Addition Polymers

Addition polymers are made by coupling the monomers by converting π bonds within each monomer to σ bonds between monomers.

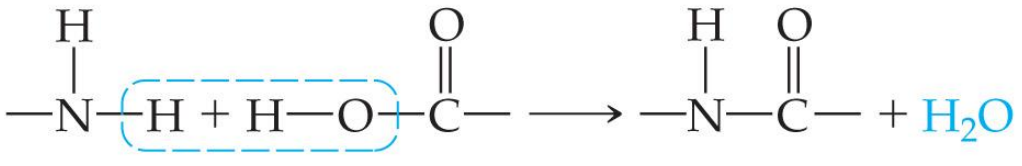


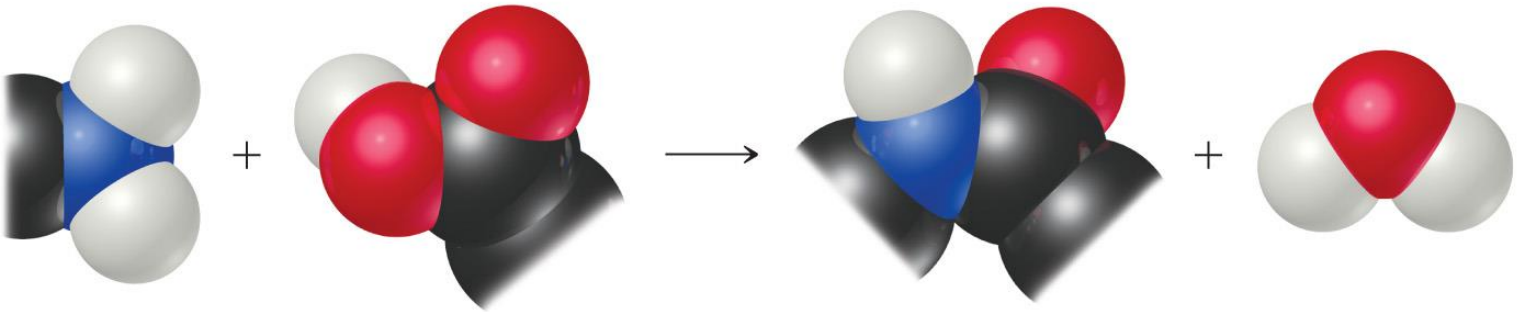
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Condensation Polymers

- **Condensation polymers** are made by joining two subunits through a reaction in which a smaller molecule (often water) is also formed as a by-product.
- These are also called **copolymers**.



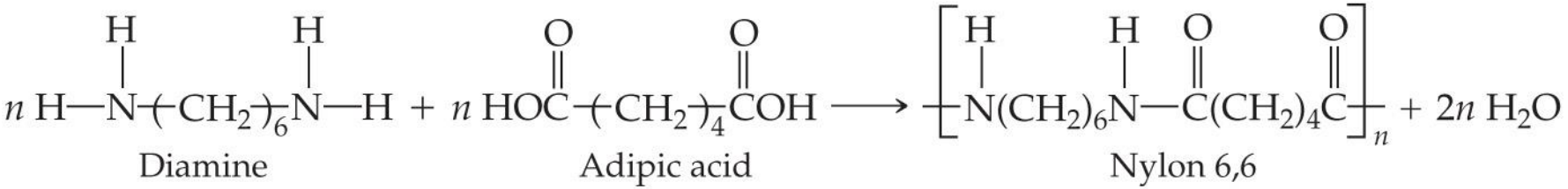


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Synthesis of Nylon



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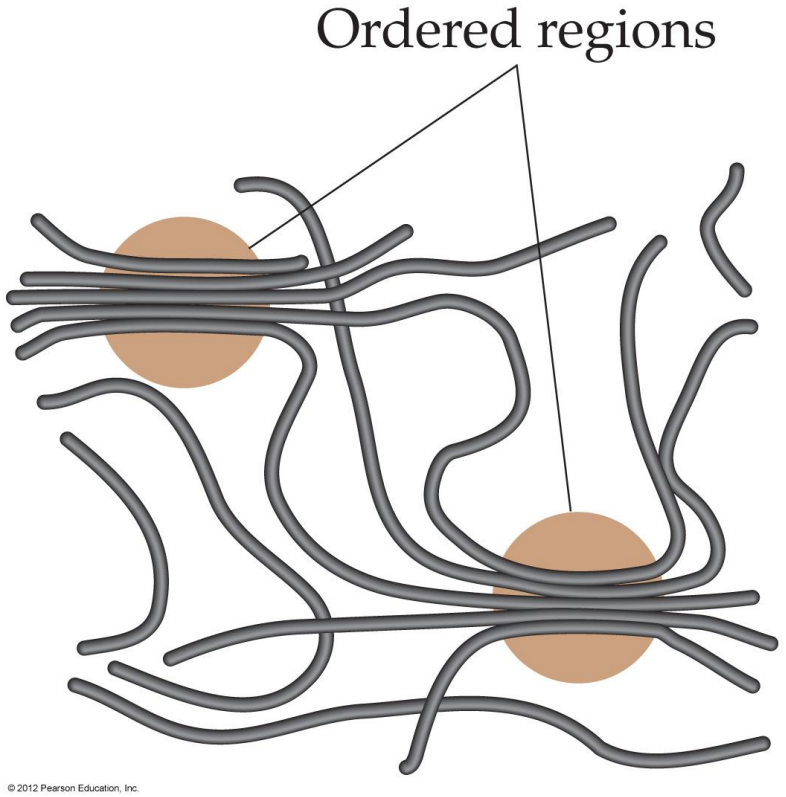
Nylon is one example of a condensation polymer.

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Properties of Polymers

Interactions between chains of a polymer lend elements of order to the structure of polymers.



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Properties of Polymers

Stretching the polymer chains as they form can increase the amount of order, leading to a degree of crystallinity of the polymer.

TABLE 12.7 • Properties of Polyethylene as a Function of Crystallinity

	Crystallinity				
	55%	62%	70%	77%	85%
Melting point (°C)	109	116	125	130	133
Density (g/cm^3)	0.92	0.93	0.94	0.95	0.96
Stiffness*	25	47	75	120	165
Yield stress*	1700	2500	3300	4200	5100

*These test results show that the mechanical strength of the polymer increases with increased crystallinity. The physical units for the stiffness test are psi $\times 10^{-3}$ (psi = pounds per square inch); those for the yield stress test are psi. Discussion of the exact meaning and significance of these tests is beyond the scope of this text.

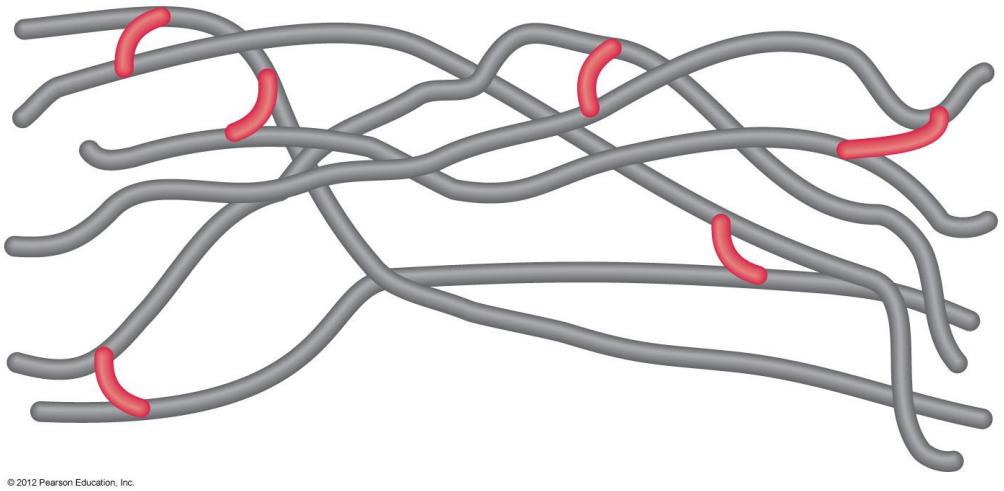
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Cross-Linking

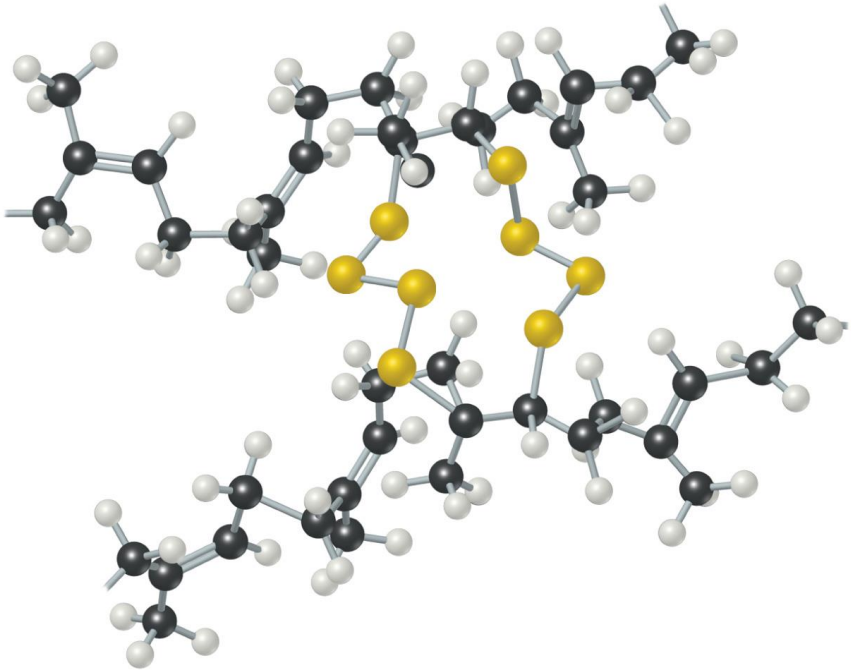
Chemically bonding chains of polymers to each other can stiffen and strengthen the substance.



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(b)

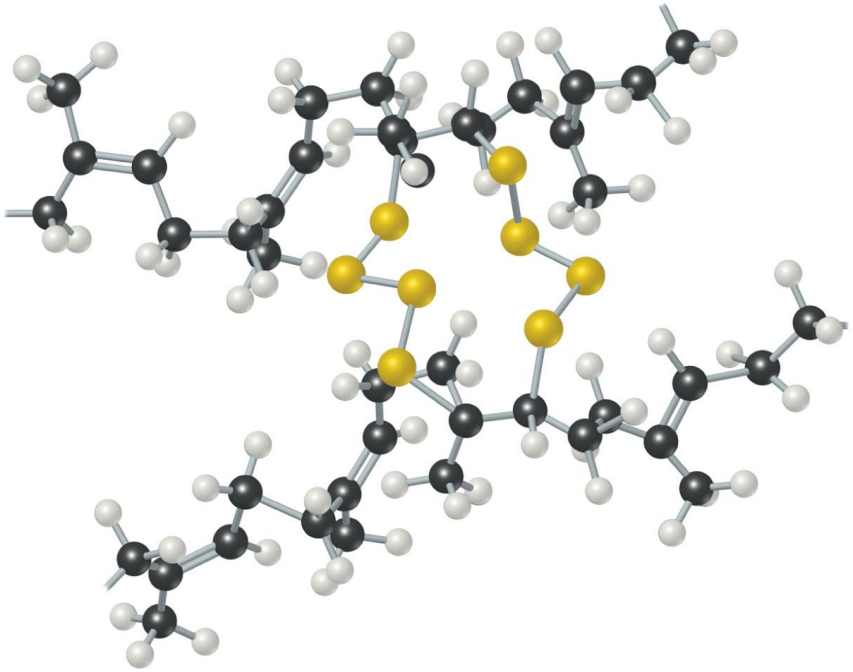
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Naturally occurring rubber is too soft and pliable for many applications.

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Cross-Linking



(b)

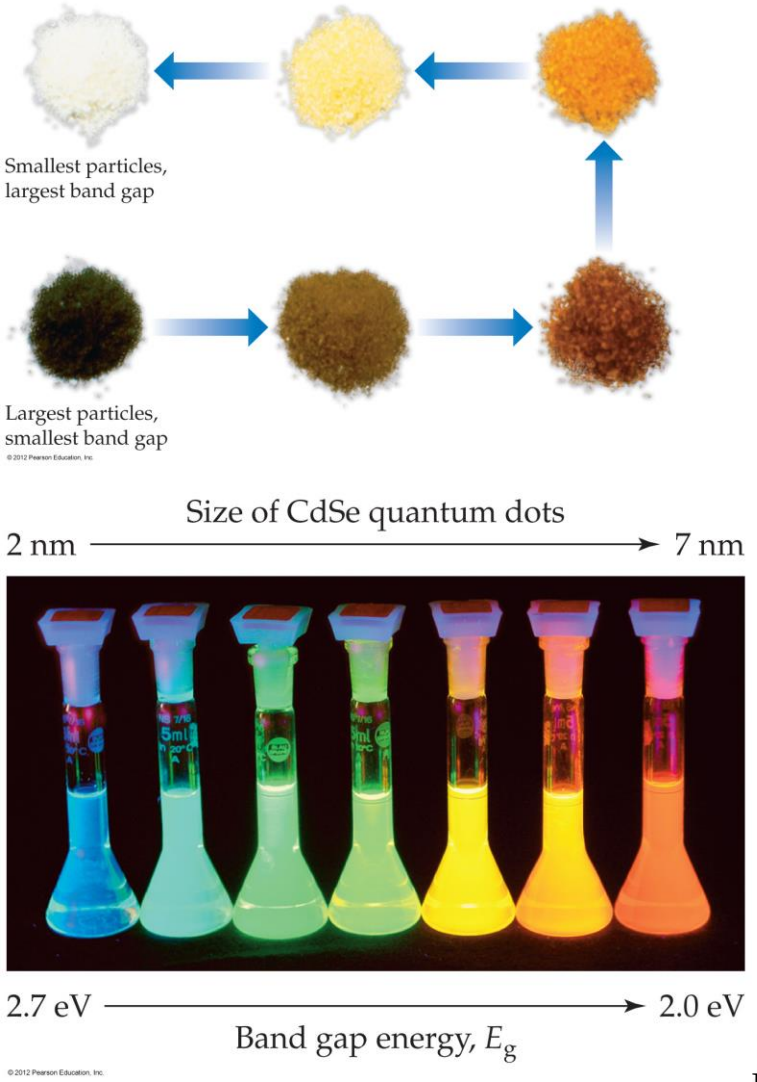
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In **vulcanization**, chains are cross-linked by short chains of sulfur atoms, making the rubber stronger and less susceptible to degradation.

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Nanoparticles

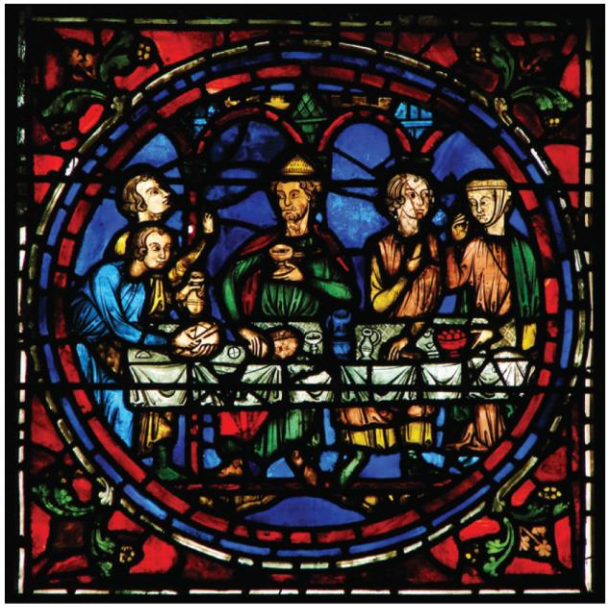
Different size particles of a semiconductor (like Cd_3P_2) can emit different wavelengths of light, depending on the size of the energy gap between bands.



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Nanoparticles



Finely divided metals can have quite different properties than larger samples of metals.



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